

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Here are 10 mental math questions based on the core concepts in the provided text:

Basic Trigonometric Equations

1. Sine Equation

- For what values of θ in $[0, 2\pi)$ is $\sin \theta = 0$?

2. Cosine Equation

- If $\cos \theta = 1$, what are all possible θ in $[0, 2\pi)$?

3. Tangent Equation

- For what values of θ in $[0, 2\pi)$ is $\tan \theta = 0$?
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Zero-Product Property

4. Using Zero-Product Property

- If $(\sin \theta)(\cos \theta) = 0$, what are the possible solutions for θ in $[0, 2\pi)$?
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Factoring Trigonometric Equations

5. Factoring & Solving

- Solve: $2 \sin \theta - 1 = 0$. What is $\sin \theta$? What is θ in $[0, 2\pi)$?

6. Quadratic in Sine

- If $\sin^2 \theta = 1$, what are the possible values of θ in $[0, 2\pi)$?
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Trigonometric Identities

7. Pythagorean Identity Application

- If $\sin \theta = 1/2$, what is $\cos^2 \theta$?
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Multiple Angles

8. Double Angle Concept

- If $\sin 2\theta = 0$, what can θ be in $[0, 2\pi)$?
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Trigonometric Manipulation

9. Equation Rearrangement

- If $\cos \theta = 2 \sin \theta$, what is $\tan \theta$?
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Solution Generalization

10. All Solutions with Periodicity

- If $\sin \theta = 1/2$, write a general formula for all possible solutions for θ .
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Let me know if you would like the answers or full steps as well!